

Project 2

ELECENG 2EI4 – March 1st, 2026

Adi Ganguli - 400572384

Statement of Originality: *I hereby certify that all the work described within this report is the original work of the author. Any published (or unpublished) ideas and/or techniques from the work of others are fully acknowledged in accordance with the standard referencing practices, and I have obtained the necessary permission from the organization to submit this report.*

Table of Contents

Switch Properties and Test Plans	3
Ideal switches: a description of the important properties of an ideal switch	3
Switch non-idealities:	3
Qualitatively describe each non-ideality that is expected in a real switch.	3
Quantitatively state the behavior caused by each non-ideality.	3
Test Plans: For each non-ideality specified above, describe the experiment that you will use to determine the quantitative performance of your design. This includes:	4
Values that you will set for VControl, VSupply, and V1.	4
Values that you measure, and values that you will calculate based on the measurement.	4
Switch 1	5
A circuit schematic of your design.	5
Measurements performed according to the test plan.	5
Circuit Simulation	6
Real-Life Circuit Test	7
Theoretical explanation for the results obtained and comparison of the quantitative results with theory.	8
Design tradeoffs. What tradeoffs did you make in your design for performance, complexity, and cost?	9
Switch 2	10
A circuit schematic of your design.	10
Measurements performed according to the test plan.	10
Circuit Simulation	11
Real-Life Circuit Test	12
Theoretical explanation for the results obtained and comparison of the quantitative results with theory.	13
Design tradeoffs. What tradeoffs did you make in your design for performance, complexity, and cost?	14

Switch Properties and Test Plans

Ideal switches: a description of the important properties of an ideal switch

Ideal switches have these important properties:

- When turned off the current running through it is zero (open circuit), and when turned on, the voltage drop across the switch is zero (short circuit). In both cases no power is dissipated throughout the switch
 - Therefore, $V_{on} = 0 \text{ V}$, $I_{off} = 0 \text{ A}$, $P = 0 \text{ W}$.
- Zero resistance when the switch is closed, and infinite resistance when open
 - Therefore, $R_{on} = 0 \Omega$, $R_{off} = \infty \Omega$.
- Allows current in both directions when on
- Switches instantaneously (current does not rise and fall between opening or closing switch)
- Voltages at each end of the switch (V_1 and V_2) can have any value, as there is no breakdown voltage of an ideal switch (i.e perfect insulator)

Switch non-idealities:

Qualitatively describe each non-ideality that is expected in a real switch.

In a non-ideal switch, there is:

- Small resistance when the switch is on, meaning:
 - A small voltage drop exists between the two ends of the switch
 - Current going through the switch is non-ideal
 - There is now power dissipated between the two ends of the switch (heat loss)
- Devices that are used to make switches (e.g MOSFETS, Zener Diode) don't shut off current completely due to imperfect induction
 - This means very small current can still pass through the switch
- When the switch goes from open to closed, there is a decrease in current rather than an instantaneous drop in current
- Real switches have maximum voltage ratings/limits. If V_1 and V_2 go past this limit, the switch will go into breakdown when it is off, causing current to conduct through it

Quantitatively state the behavior caused by each non-ideality.

- Small resistance when the switch is on, meaning:
 - $V_{drop} = v_1 - v_2 = i \times R_{internal}$
 - $P_{loss} = i^2 R_{internal}$
 - $V_1 > V_{min}$ and $V_2 < V_{max}$ to prevent breakdown

Test Plans: For each non-ideality specified above, describe the experiment that you will use to determine the quantitative performance of your design. This includes:

Values that you will set for V_{Control} , V_{Supply} , and V_1 .

For both switches:

- $V_{\text{Supply}} = V_1 = 5 \text{ V}$.
- V_{Control} is a square wave oscillating from 0 V to 5 V with a frequency of 500 mHz.

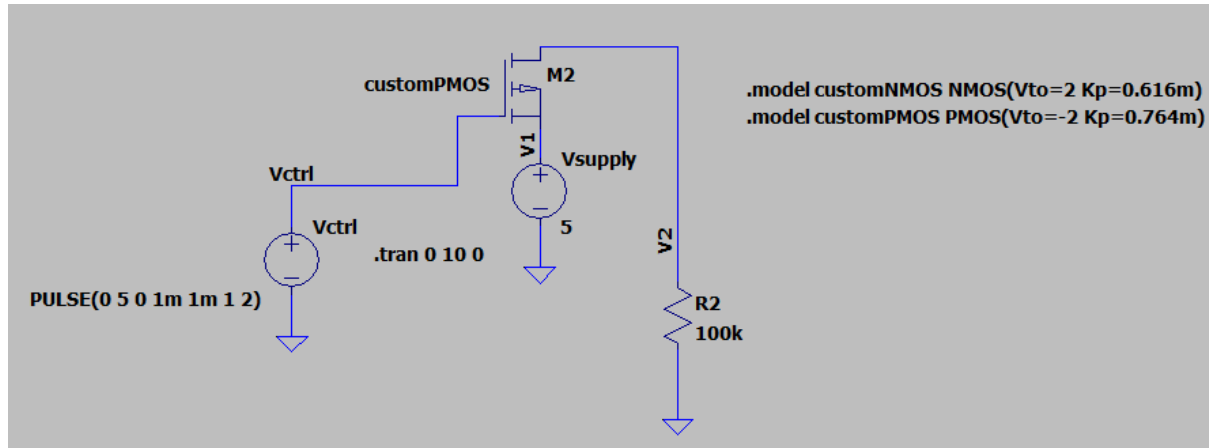
Values that you measure, and values that you will calculate based on the measurement.

The biggest non-ideality that will happen with the switch designs is that they have a small internal resistance. This causes voltage drop across the switches, and a small leakage current. To measure both of these, the voltage drop and leakage current will be measured using the AD3 voltmeter. Leakage current will require a load resistor since the AD3 does not have an ammeter. These factors confirm that the switch design is not completely bidirectional. To see this, R_{ON} can be determined with Ohm's law and its relationship with the leakage current and voltage drop across the switch, specifically comparing the currents on each end of the MOSFET design (i.e. i_1 and i_2).

Switch 1

A circuit schematic of your design.

Figure 1: Circuit Schematic of Switch 1 in LTSpice



Measurements performed according to the test plan.

Using the IV characteristics of the MOSFET measured with the AD3, the K and V_T are:

NMOS

$$V_{TN} = 2.0 \text{ V}$$

$$K_N = 0.616 \text{ mA/V}^2$$

PMOS

$$V_{TP} = -2.0 \text{ V}$$

$$K_P = 0.764 \text{ mA/V}^2$$

Theoretical Calculations

Voltage Drop when ON $V_{\text{Control}} = 0 \text{ V}$	For the PMOS, assume Saturation: $V_{SD_1} \geq (V_{SG_1} + V_{TP})$ $V_{SD_1} \geq (5V - 2.0V)$ $5.0V \geq 3.0V$ $V_{\text{Drop}} = V_2 - V_1 = 5.0V - 5.0V = 0V$ This is the ideal drop, but in the experiment
---	--

	this will be nonzero.
Current Leakage when OFF $V_{\text{Control}} = 5 \text{ V}$	For the PMOS, check if in cutoff: $V_{SG_1} \leq V_{TP}$ $5\text{V} - 5\text{V} \leq 2.0\text{V}$ $0\text{V} \leq 2.0\text{V}$ Therefore, the PMOS goes into cutoff with this voltage, and $i_D = 0 \text{ mA}$. When the actual switch is tested, the leakage current is the current running through the load resistor R_L. $i_{\text{Leak}} = \frac{V_2}{R_L}$
R_{ON} calculation when ON (bidirectional) $V_{\text{Control}} = 0 \text{ V}$	For the PMOS, we know it is in Saturation: $R_{\text{ON (forward)}} = \frac{v_1 - v_2}{i_2}$ $R_{\text{ON (backward)}} = \frac{v_2 - v_1}{i_1}$ In theory, the current in both states will be equal, but in the experiment, the current for both states will be examined at the V_2 and V_1 nodes for forward and backward resistances respectively. Ideally, $R_{\text{ON (forward)}} = R_{\text{ON (backward)}} = 0 \Omega$

Circuit Simulation

Figure 2: Relationship between V_2 (green) and V_{Control} (blue), V_{Control} closes the switch at 0V and opens at 5V , and V_2 is either 5V or 0V respectively, meaning the switch is working

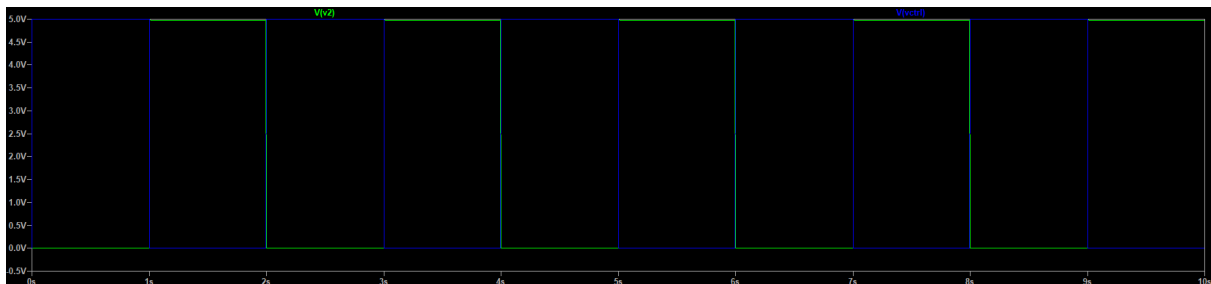
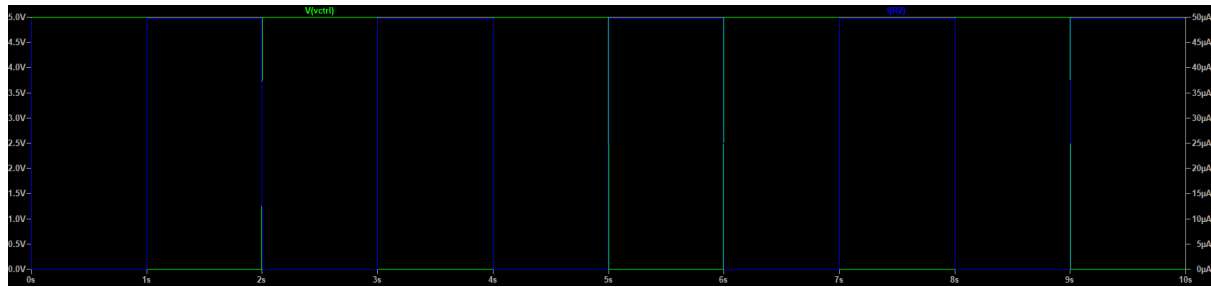


Figure 3: Relationship between i_D (blue) and $V_{Control}$ (green), similar relationship to Figure 3, which indicates that the switch is working as intended



Real-Life Circuit Test

Figures 4 & 5: Circuit Design on the Breadboard using the AD3

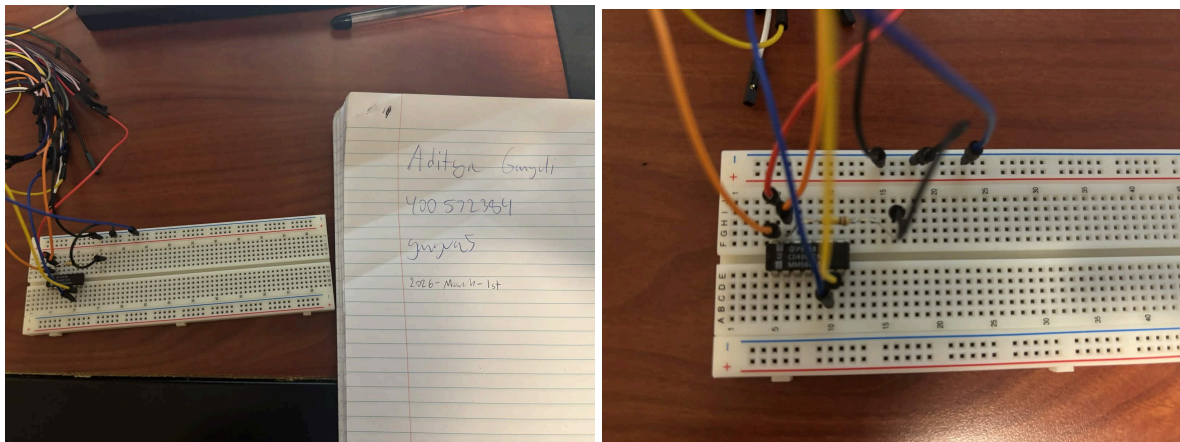


Figure 6: $V_{Control}$ measured with AD3. $V_{Control}$ oscillates between 31 mV (on) and 5.025 V (off)

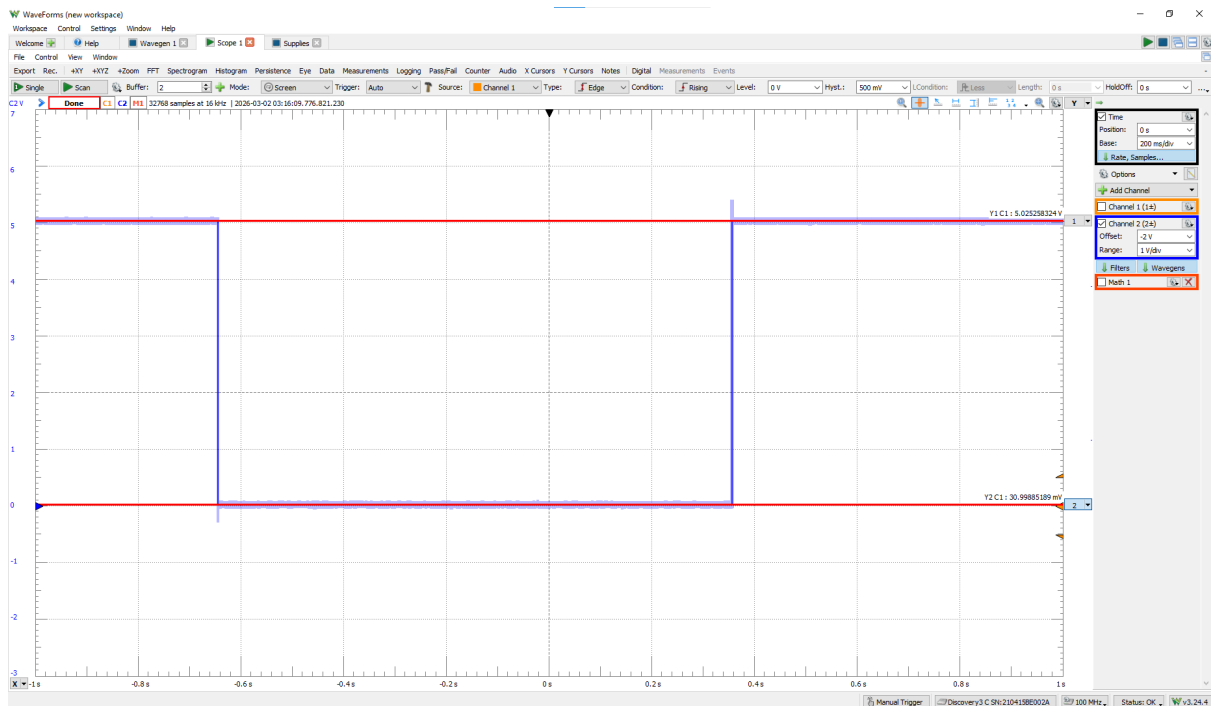
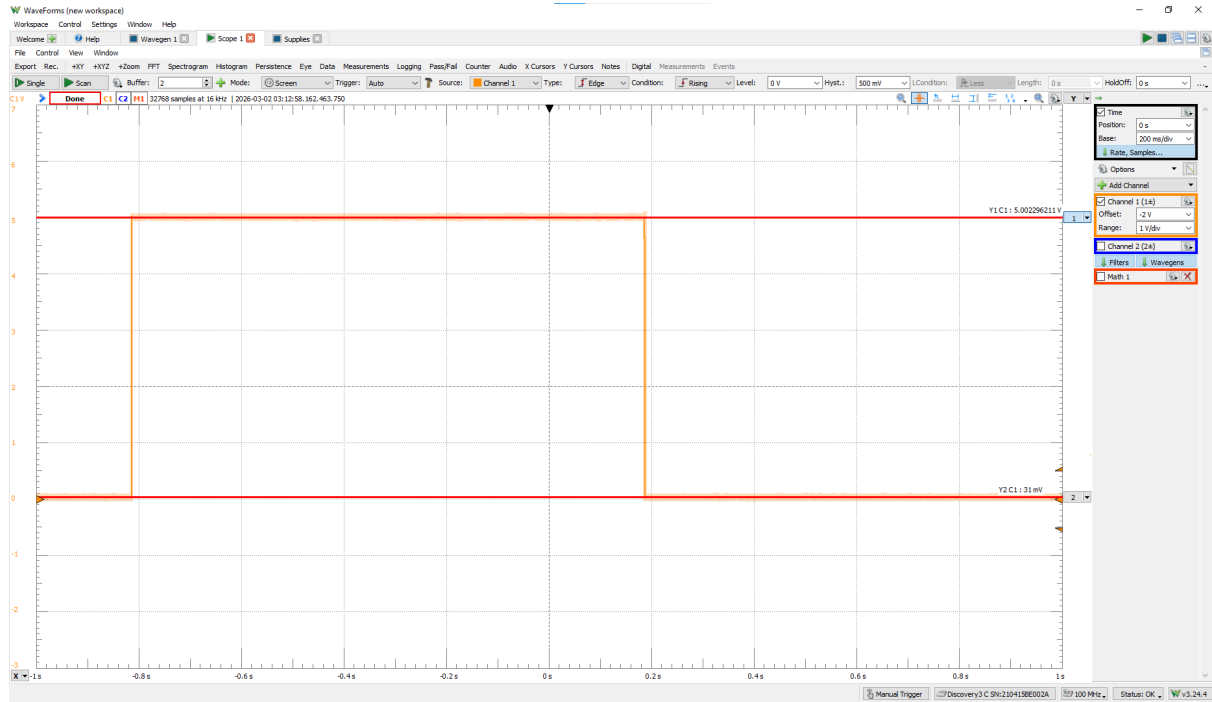


Figure 7: V_2 measured with the AD3, $V_{2, on} = 5.003V$, $V_{2, off} = 31 mV$ 

Theoretical explanation for the results obtained and comparison of the quantitative results with theory.

Voltage Drop when ON $V_{Control} = 0 V$	$V_{Drop} = V_2 - V_1$ $V_{Drop} = 5.002V - 5.025V = -0.022V$
Current Leakage when OFF $V_{Control} = 5 V$	$i_{Leak} = \frac{V_2}{R_L} = \frac{31 mV}{100 k\Omega} = 0.31 \mu A$
R_{ON} calculation when ON (bidirectional) $V_{Control} = 0 V$	Measuring from AD3: $i_1 = 50.80 \mu A$, $i_2 = 50.22 \mu A$ $R_{ON (forward)} = \frac{v_1 - v_2}{i_2} = \frac{0.022V}{50.22 \mu A} = 438 \Omega$ $R_{ON (backward)} = \frac{v_2 - v_1}{i_1} = \frac{0.022V}{50.80 \mu A} = 433 \Omega$

After building and testing the circuit using the AD3, it can be seen that the measured non-idealities are quite small in testing. The voltage drop between the switch when ON resulted in a drop of 0.022 V, the leakage current when OFF is measured to be 0.31 μA through the small voltage at node V_2 when off and the load resistor R_L . Finally, it is seen that

R_{ON} is not the same in both directions, with the internal resistance being $438\ \Omega$ forward, and $433\ \Omega$ backward. These values are very small, with the combination of the voltage drop, current leakage and R_{ON} values showing that this switch is almost ideal, which proves that the design is reasonably scalable and meets expectations.

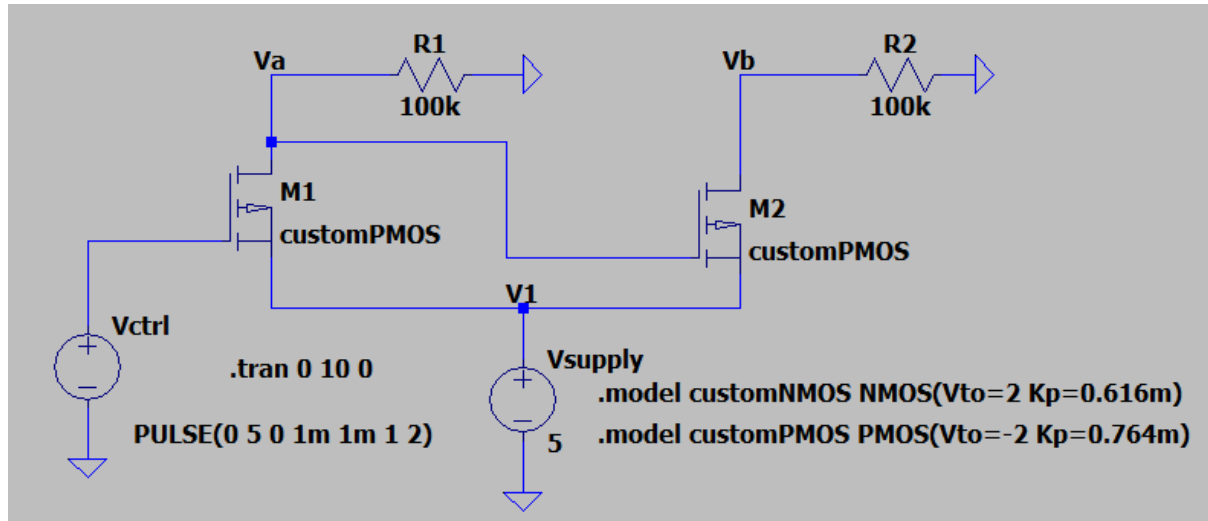
Design tradeoffs. What tradeoffs did you make in your design for performance, complexity, and cost?

As the function of this switch is very simple, the chosen design has very little complexity and performs well, as seen with the accurate voltage transfer between the PMOS. One CD4007CN chip is used with just the PMOS that has its bulk already shorted, making it very simple to use along with the resistors and wiring. One major tradeoff with this design (and switch 2) is that the voltage drop is dependent on the load resistance. I found this out when using a different load resistor (e.g $330\ \Omega$) to test the value of i_D , since a value of $100\ k\Omega$ will force the theoretical current of $\sim 30\ mA$ to be $50\ \mu A$ in order to achieve a V_2 of $5\ V$. I consider this a tradeoff as an ideal switch would have both have $V_1 = V_2$ (i.e. a short circuit) and the current not being forced into a low value. Drawing more current from the PMOS would create a large voltage drop across its internal resistance R_{ON} , which is a limitation of this design. As for the cost, this simple build is extremely cheap, only requiring one CD4007CN IC chip, a resistor and basic wires.

Switch 2

A circuit schematic of your design.

Figure 8: Circuit Schematic of Switch 2 in LTSpice



Measurements performed according to the test plan.

Using the IV characteristics of the MOSFET measured with the AD3, the K and V_T are:

NMOS

$$V_{TN} = 2.0 \text{ V}$$

$$K_N = 0.616 \text{ mA/V}^2$$

PMOS

$$V_{TP} = -2.0 \text{ V}$$

$$K_P = 0.764 \text{ mA/V}^2$$

Theoretical Calculations

Note that much of the theory is similar to the work done in Switch 1:

Voltage Drop when $V_1 = V_A$ $V_{\text{Control}} = 0 \text{ V}$	$V_{\text{Drop, A}} = V_1 - V_A$ $V_{\text{Drop, A}} = 5.0\text{V} - V_A$ $V_{\text{Drop, A}} = 0 \text{ V ideally, but this will be nonzero when tested}$
Voltage Drop when $V_1 = V_B$	$V_{\text{Drop, B}} = V_1 - V_B$

$V_{\text{Control}} = 5 \text{ V}$	$V_{\text{Drop, B}} = 5.0\text{V} - V_{\text{B}}$ $V_{\text{Drop, B}} = 0 \text{ V}$ ideally, but this will be nonzero when tested
Current Leakage when $V_1 = V_A$ OFF $V_{\text{Control}} = 5 \text{ V}$	$i_{\text{Leak, A}} = \frac{V_A}{R_L}$ Similar to the voltage drops, $i_{\text{Leak, A}} = 0 \text{ A}$ ideally.
Current Leakage when $V_1 = V_B$ OFF $V_{\text{Control}} = 0 \text{ V}$	$i_{\text{Leak, B}} = \frac{V_B}{R_L}$ Similar to the voltage drops, $i_{\text{Leak, B}} = 0 \text{ A}$ ideally.
R_{ON} calculation when ON (bidirectional) $V_{\text{Control}} = 0 \text{ V}$	$R_{\text{ON (forward)}} = \frac{v_1 - v_{A/B}}{i_{1, A/B}}$ $R_{\text{ON (backward)}} = \frac{v_{A/B} - v_1}{i_{2, A/B}}$ Similar to the calculation in Switch 1. This will apply to both the V_A and V_B set of MOSFET's. Ideally, $R_{\text{ON (forward)}} = R_{\text{ON (backward)}} = 0 \Omega$

Circuit Simulation

Figure 9: Relationship between V_A (blue) and V_{Control} (green), with $V_1 = V_{\text{Supply}} = 5 \text{ V}$. V_{Control} sets V_A to 5 V when at 0 V , and sets V_A to 0 V when at 5 V meaning the switch is working as intended.

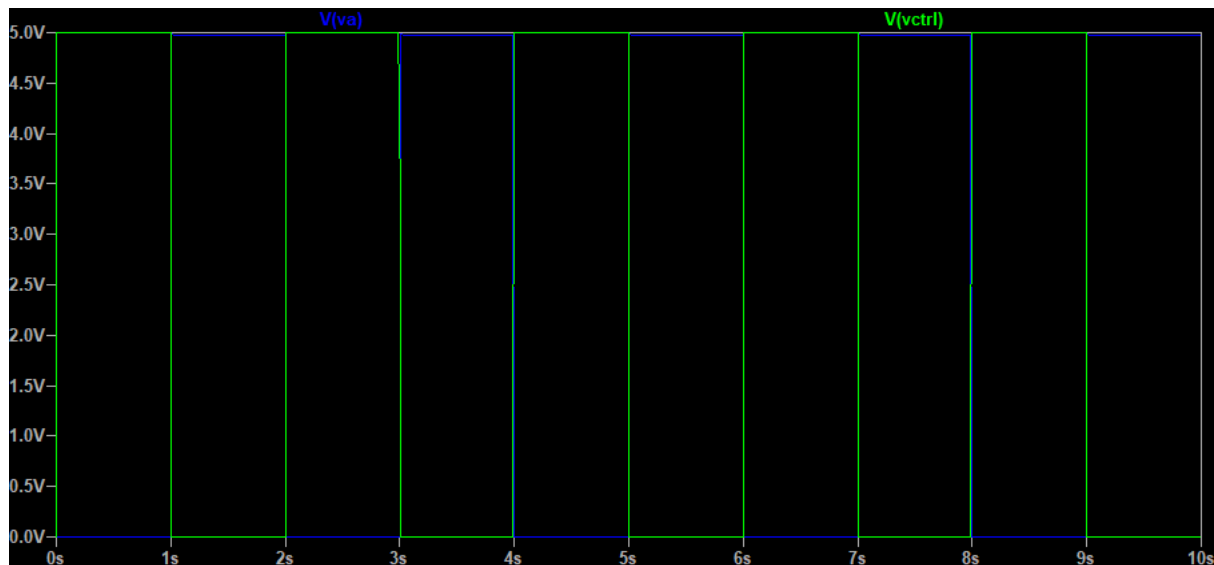
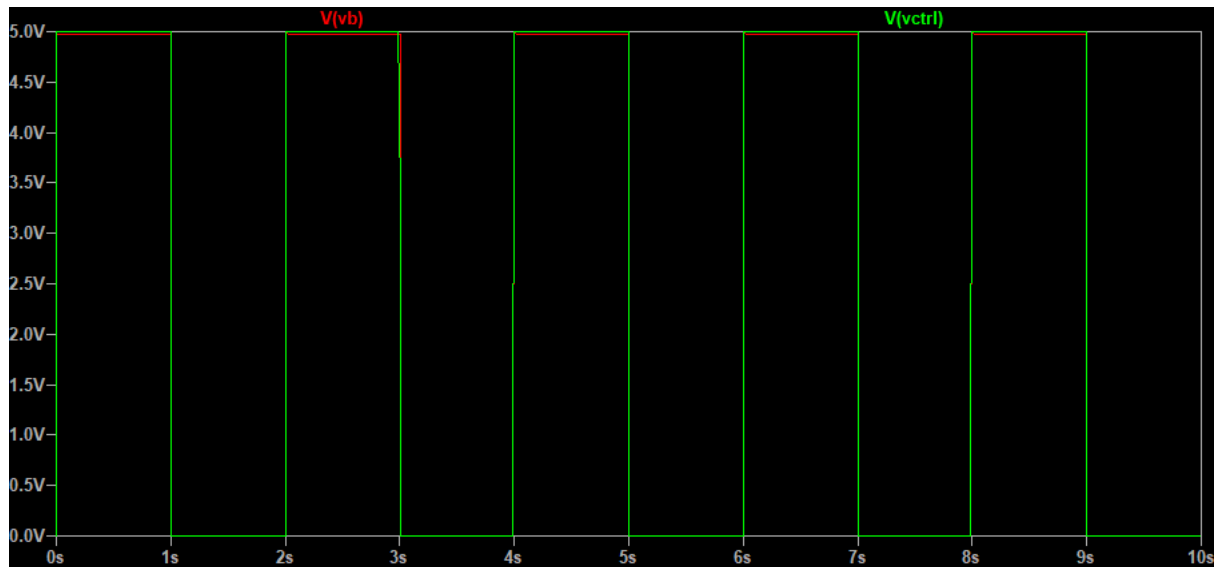


Figure 10: Relationship between V_B (red) and $V_{Control}$ (green), with $V_I = V_{Supply} = 5\text{ V}$. $V_{Control}$ sets V_A to 5 V when at 0 V , and sets V_A to 0 V when at 5 V meaning the switch is working as intended.



Real-Life Circuit Test

Figures 11 & 12: Circuit Design on the Breadboard using the AD3

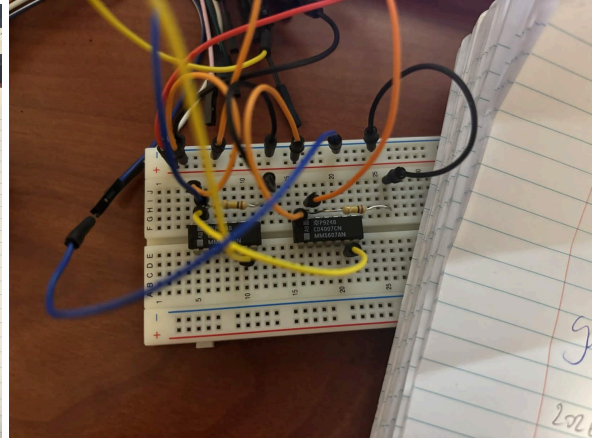
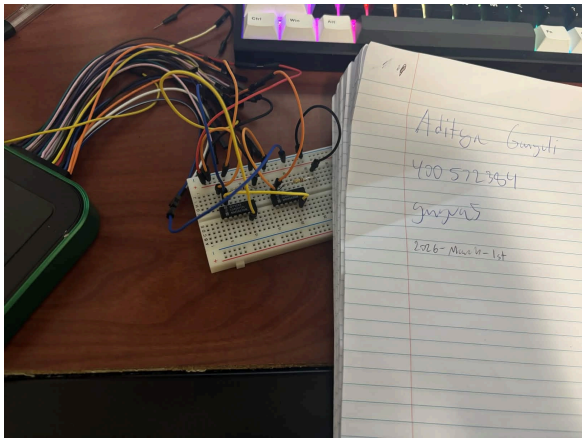


Figure 13: Relationship between V_A (blue) and $V_{Control}$ (orange), on the AD3 with $V_I = V_{Supply} = 5\text{ V}$. $V_A = 5.002\text{ V}$ when $V_{Control} = 31\text{ mV}$, and $V_A = 31\text{ mV}$ when $V_{Control} = 5.025\text{ V}$

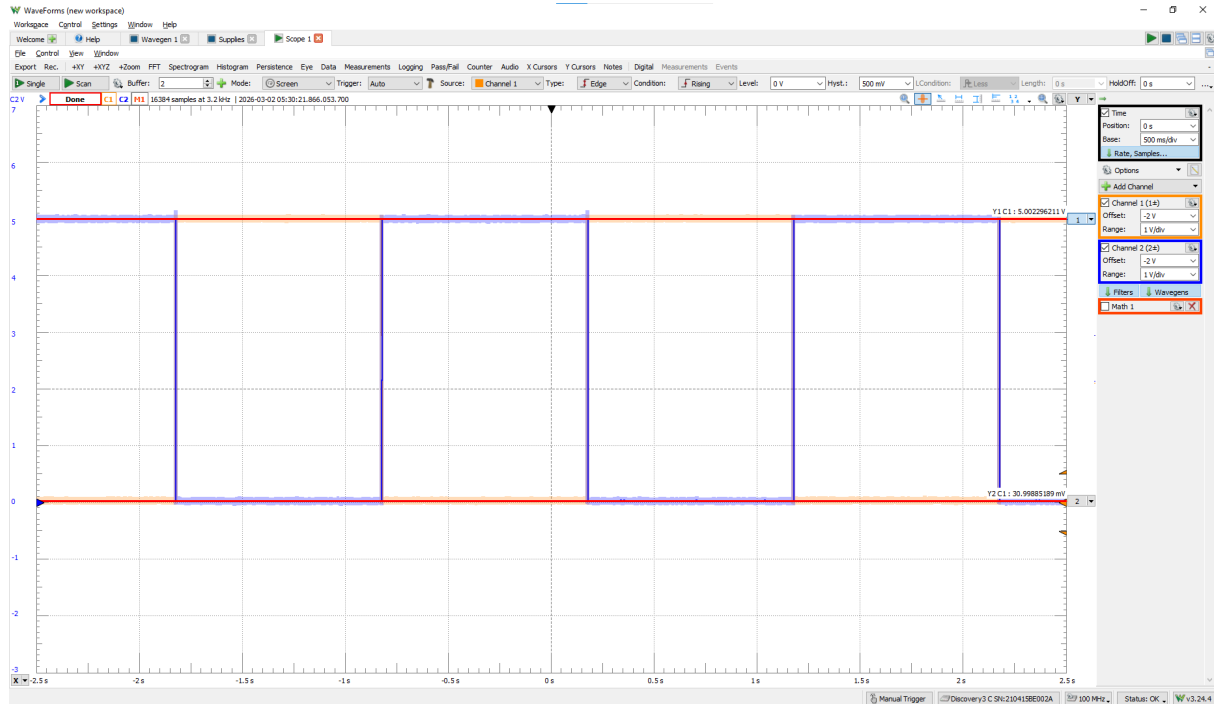
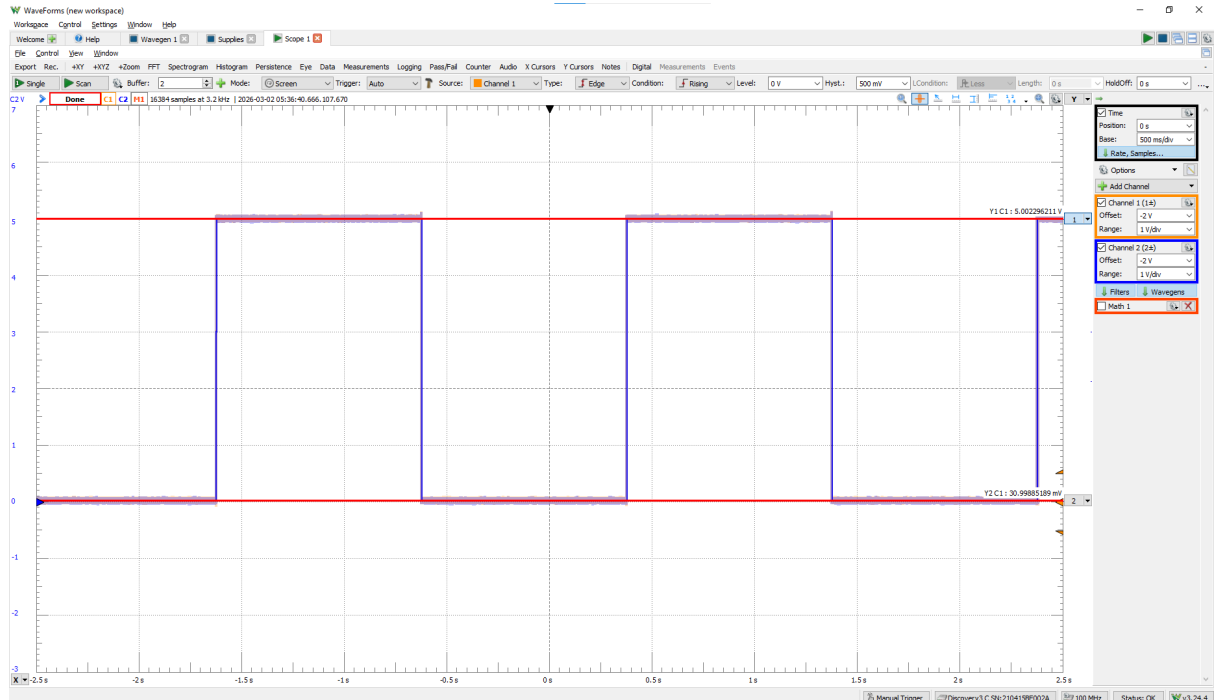


Figure 14: Relationship between V_B (orange) and $V_{Control}$ (blue), on the AD3 with $V_I = V_{Supply} = 5\text{ V}$. $V_B = 31\text{ mV}$ when $V_{Control} = 31\text{ mV}$, and $V_B = 5.002\text{ mV}$ when $V_{Control} = 5.025\text{ V}$



Theoretical explanation for the results obtained and comparison of the quantitative results with theory.

Voltage Drop when $V_1 = V_A$ $V_{\text{Control}} = 0 \text{ V}$	$V_{\text{Drop}, A} = V_1 - V_A$ $V_{\text{Drop}, A} = 5.025 \text{ V} - 5.002 \text{ V} = 0.022 \text{ V}$
Voltage Drop when $V_1 = V_B$ $V_{\text{Control}} = 5 \text{ V}$	$V_{\text{Drop}, B} = V_1 - V_B$ $V_{\text{Drop}, B} = 5.025 \text{ V} - 5.002 \text{ V} = 0.022 \text{ V}$
Current Leakage when $V_1 = V_A$ OFF $V_{\text{Control}} = 5 \text{ V}$	$i_{\text{Leak}, A} = \frac{V_A}{R_L} = \frac{31 \text{ mV}}{100 \text{ k}\Omega} = 0.31 \mu\text{A}$
Current Leakage when $V_1 = V_B$ OFF $V_{\text{Control}} = 0 \text{ V}$	$i_{\text{Leak}, B} = \frac{V_B}{R_L} = \frac{31 \text{ mV}}{100 \text{ k}\Omega} = 0.31 \mu\text{A}$
R_{ON} calculation when ON (bidirectional) $V_{\text{Control}} = 0 \text{ V}$	Measured from the AD3: $i_{1,A} = 50.63 \mu\text{A}$, $i_{2,A} = 50.17 \mu\text{A}$, $i_{1,B} = 50.40 \mu\text{A}$, $i_{2,B} = 49.94 \mu\text{A}$ $R_{\text{ON (forward), A}} = \frac{v_1 - v_A}{i_{1,A}} = \frac{0.022 \text{ V}}{50.63 \mu\text{A}} = 434 \Omega$ $R_{\text{ON (backward), A}} = \frac{v_A - v_1}{i_{2,A}} = \frac{0.022 \text{ V}}{50.17 \mu\text{A}} = 438 \Omega$ $R_{\text{ON (forward), B}} = \frac{v_1 - v_B}{i_{1,B}} = \frac{0.022 \text{ V}}{50.40 \mu\text{A}} = 436 \Omega$ $R_{\text{ON (backward), B}} = \frac{v_B - v_1}{i_{2,B}} = \frac{0.022 \text{ V}}{49.94 \mu\text{A}} = 440 \Omega$

Even though the design is generally more complex, the values of the voltage drops and current leakages are identical for V_A , V_B and the Switch 1 design, most likely due to how the design for Switch 2 is built off of Switch 1, showing that Switch 2 also faces the same issues, albeit on the same scale. The R_{ON} values are slightly different from Switch 1 when the current was measured with the AD3, as there are many more connections with 2 PMOS being used in the design, leading to more unwanted electrical noise or internal resistances from the PMOS and breadboard. This switch design is close to an ideal design, and generally meets expectations as the non-idealities meet the expected ranges of the switch.

Design tradeoffs. What tradeoffs did you make in your design for performance, complexity, and cost?

This design was relatively simple for a more complicated concept, as it mimics the system of a single-pole double-throw switch (SPDT). The use of two PMOS made it simple to use, but I used two CD4007CN chips to use both PMOS that had the bulk shorted out for simplicity, so I could have used one chip instead. The most important thing to note in this design is that the voltage switch is very accurate due to the high load resistor value, which is similar to Switch 1. This means this design would generally fail without it. I found this through testing the

voltage drop with a lower resistor (e.g $330\ \Omega$), and noticed a significantly larger voltage drop. Therefore, while this design functions as a voltage controlled SPDT, a high resistance is required at the drain of both MOSFET's to get a good voltage transfer. As for cost, this is an ideal build for a cheap SPDT, as the IC, wires and resistors cost very little.